"MORE THAN WORDS": GENRE-DISCRIMINATING LYRICAL PATTERNS IN 50 YEARS OF ANGLOPHONE GERMAN POPULAR MUSIC

Anonymous Authors
Anonymous Affiliations
anonymous@ismir.net

ABSTRACT

The abstract should be placed at the top left column and should contain about 150–200 words.

Warum nciht textuelle Statistik; wie werden topic models und sentimente in der KoWi gefasst (nomen, Adjektive); was ist idiomatic expressions

1. INTRODUCTION

RQ1: What are content-related genre-discriminative lyrical patterns (topics, sentiments, idiomatic expressions) in Anglophone German popular music? - Dictionaries Figthing Words - LIFT Words for Topics, Sentiments, and Expressions (in Appendix)

RQ2: How well do interpretative ML-models perform when predicting popular music genres compared to a state of the art tranformer-based LLM? - ML table - Confusion matrix for reduced Model - Permutation importance for Combined (Appendix)

RQ3: What can we learn about lyrical genre differences from such interpretative models for popular music studies? - Permutation Importance (Transition, show Distilled) - Cluster Probs per genre - SHAP for Rock

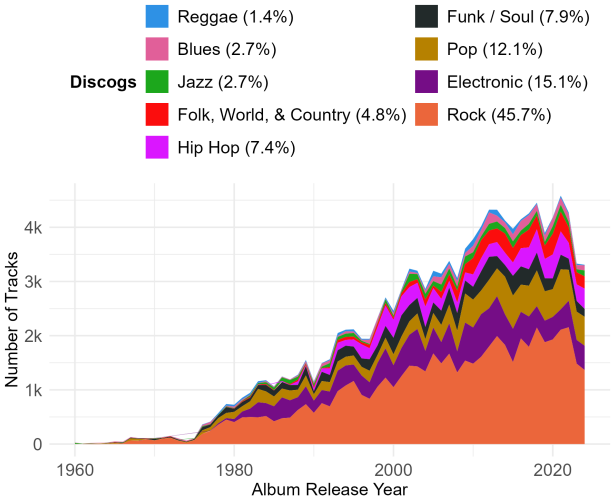

Figure 1: Composition of the analyzed corpus.

2. METHODS

2.1 Data

gemeinsames Metadaten-Framework für Extension und Anbindung an Audio-Features

VOrgefertigter Datnsatz

Corpus rechtfertigung:

Es geht um Musikhören, nicht nur um Verkäufe; auch Musik, die schon länger nicht mehr in den Charts ist, daher anreiherung;

Single Label Discogs Genres: im Lyrics Paper würde ich begründen mit Interpretierbarkeit

2.2 Genre Disrciminating Vocabularies

$$E = mc^2 \tag{1}$$

2.3 Genre Discriminating Topics and Styles

2.4 Machine Learning Experiments

© Ananymous Authors. Licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0). **Attribution:** Ananymous Authors, “"More than words": Genre-Discriminating Lyrical Patterns in 50 Years of Anglophone German Popular Music”, submitted to *ISMIR*, 2026.

Features	Learner	Params	$F1_{\text{macro}}$
Raw Lyrics	Random		0.111
	DistillBERT [1]		0.374
TF-IDF [2]	Logistic Regression	C: 1 L1: 0.5	0.249
Topics			0.261
Sentiments			0.187
Expressions			?
Combined			?
Combined	LightGBM [3]	tuned [4]	0.312
Distilled			?

Table 1: $F1_{\text{macro}}$ Holdout evaluation for classifcation experiments. See Appendix for tuned hyperparameters.

3. RESULTS

3.1 Genre Disrciminating Vocabularies

3.2 Genre Discriminating Topics and Styles

3.3 Machine Learning Experiments

4. DISCUSSION

Multilabel Genres

	Blues	Electronic	Folk, World, & Country	Funk / Soul	Hip Hop	Jazz	Pop	Reggae	Rock
Sex	9	7	5	18	10	7	7	10	4
Scatting	1	1	1	1	1	1	1	1	1
Romantic Love	15	17	13	22	5	21	18	15	9
Resignation	9	16	10	10	5	10	14	10	17
Rastafarian identity	0	1	1	1	1	0	0	18	0
Nostalgia	24	4	13	5	3	7	6	5	7
Nightlife	5	10	4	6	4	5	6	7	5
Nature Reverie	9	12	17	5	2	11	11	5	12
Mythology / Fantasy	3	3	4	1	1	2	2	2	9
Melodics	1	1	1	1	1	1	1	1	1
Internal Conflict	10	13	13	14	11	9	17	8	12
Gore	2	4	3	2	3	2	2	3	11
Everyday Narrative	6	6	9	4	6	7	7	7	7
Christmas	3	2	4	2	1	11	3	3	1
Braggadocio	1	1	1	4	31	1	1	2	1
Black Culture	2	3	1	3	14	3	2	4	2

(a) topics

	Blues	Electronic	Folk, World, & Country	Funk / Soul	Hip Hop	Jazz	Latin	Pop	Reggae	Rock
Weathered / Spritual	31	3	10	5	2	7	8	5	5	7
Urban / Street	2	2	2	4	24	3	4	2	5	2
Urban / Aggressive	1	1	1	8	34	1	3	1	2	1
Uncertain / Restless	11	11	11	9	3	11	9	14	10	11
Somber / Introspective	6	13	11	6	3	8	6	11	6	14
Romantic / Tender	11	13	7	23	5	13	18	13	11	5
Rastafarian	3	7	2	8	6	4	11	4	23	3
Oh / ooh / la	2	3	2	5	3	2	5	4	3	2
Nostalgic	10	10	20	5	5	12	8	12	9	11
Na na na	1	1	1	1	1	1	1	1	2	1
Festive / Yuletide	3	2	5	2	0	18	2	4	2	1
Extremely Violent	2	5	2	2	3	2	2	3	5	12
Etheral / Dreamy	7	14	10	6	2	9	9	10	7	9
Epic / Mythic	3	4	7	1	1	3	2	3	4	14
Conversational	6	10	9	14	9	6	11	15	6	8

(b) styles

Figure 2: Latent genre-discriminative topics, sentiments, and styles: probabilities per genre (TODO ACHSEN TAUSCHEN) [%].

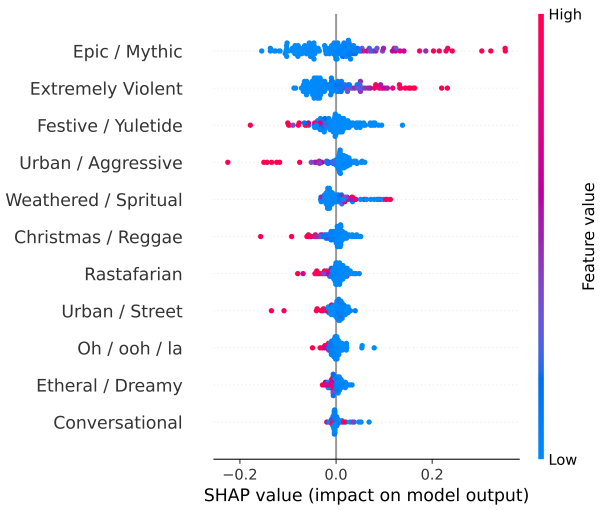

Figure 3: *Discogs' Rock* Example: SHAP values (150 samples from the holdout set) LightGBM STM Reduced.

5. AI USAGE STATEMENT

See submission guidelines. Does not count towards the page limit.

6. REFERENCES

[1] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," in *5th Workshop on Energy Efficient Machine Learning and Cognitive Computing - NeurIPS 2019*. arXiv, Mar. 2020.

[2] M. Fell and C. Sporleder, "Lyrics-based analysis and classification of music," in *Proceedings of COLING 2014, the 25th International Conference on Computational Linguistics: Technical Papers*, 2014, pp. 620–631.

[3] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Advances in Neural Information Processing Systems*, vol. 30. Curran Associates, Inc., 2017.

[4] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019.